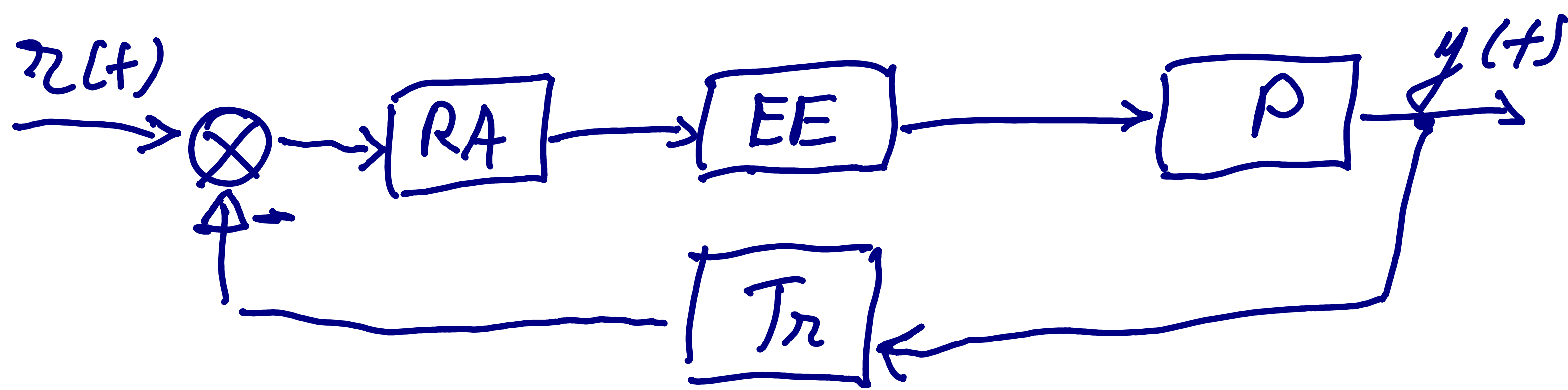


Răspunsul unui sistem

17.11.2025



$$G_e(s) = \frac{Y(s)}{R(s)}, \quad y(t) = ?$$

$$Y(s) = G_e(s) R(s) \quad | \quad \mathcal{L}^{-1} \Rightarrow y(t) = ?$$

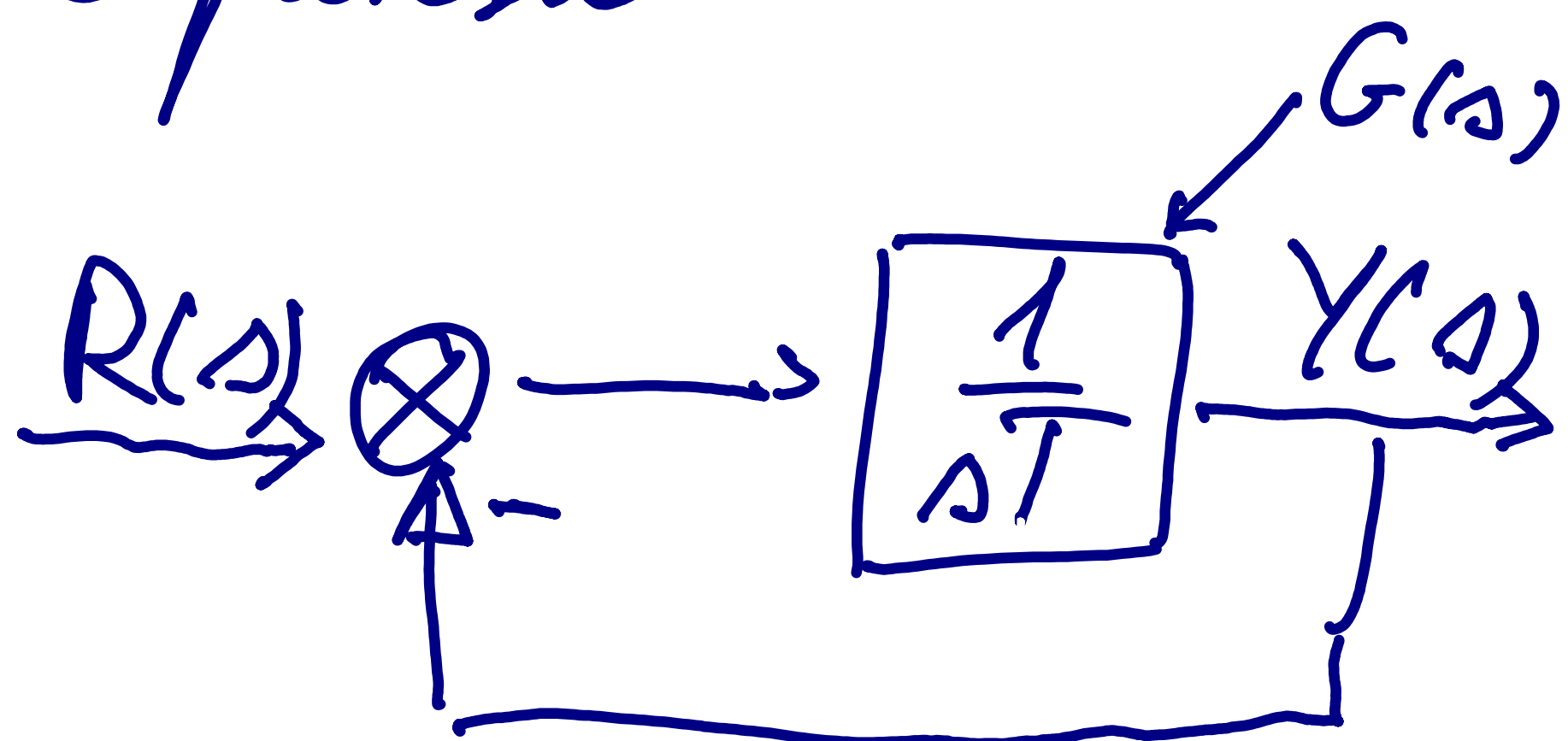
În prima etapă de dezvoltare a unui sistem de regulare este necesar să se deducă un model matematic pentru sistemul reglat.

După obt. acestui model, avem la disp. diverse metode de analiză a performanțelor sistemului. În analiză și comp. sis. trebuie să avem o bază comună pentru comparația perf. realizate de acesta în diverse situații.

$$y(t) = \underbrace{y_{st}(t)}_{y_p(t)} + y_t(t) \rightarrow \text{depinde de polii sistemului}$$

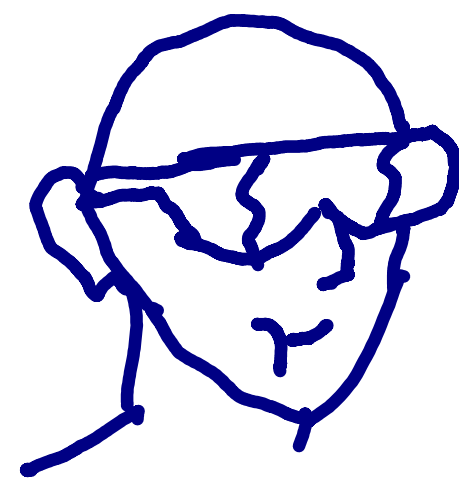
→ dat de mărimea de intrare

Răspunsul sistemului de ord. I:



$$G_e(s) = \frac{G(s)}{1+G(s)} = \frac{1}{sT+1}$$

$$G_e(s) = \frac{K}{1+G(s)} = \frac{Y(s)}{R(s)}$$



$$\begin{aligned} y(t) &= \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\{G_e(s) R(s)\} = \mathcal{L}^{-1}\left\{\frac{K}{sT+1} \cdot \frac{1}{s}\right\} = \\ &= \mathcal{L}^{-1}\left\{\frac{C_1}{s} + \frac{C_2}{sT+1}\right\} = \mathcal{L}^{-1}\left\{\frac{K}{s} - \frac{KT}{sT+1}\right\} = K - Ke^{-\frac{1}{T}t}; t \geq 0 \end{aligned}$$

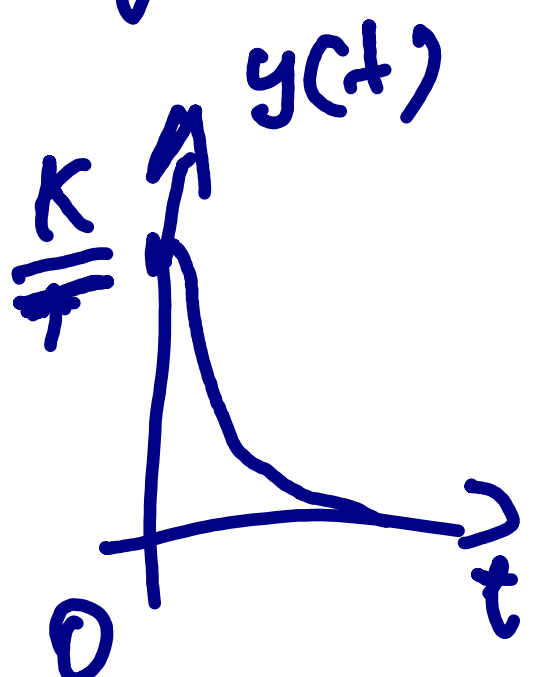
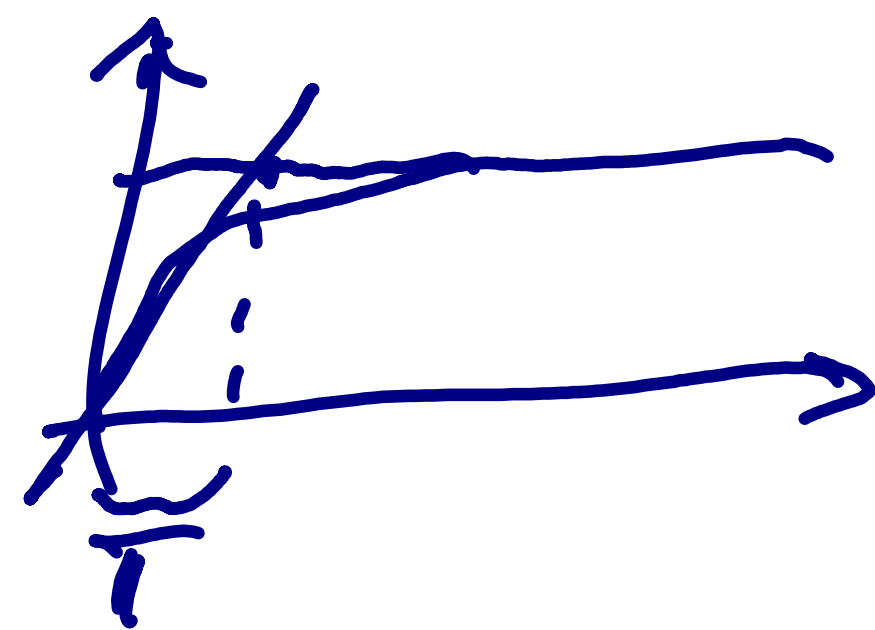
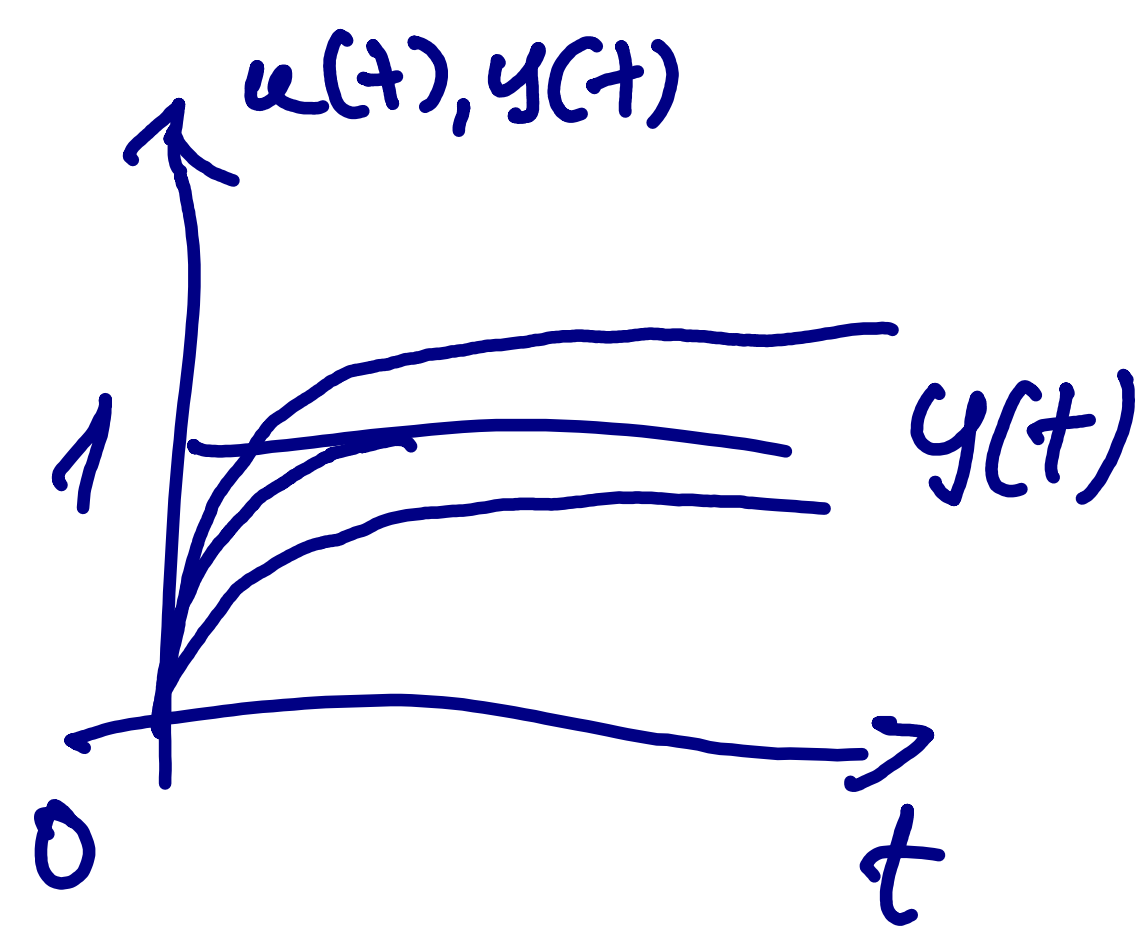
$$G_e(s) = \frac{1}{s+2} = \frac{1}{2(\frac{s}{2}+1)}$$

$$T = \frac{1}{2}$$

$$K = \frac{1}{2}$$

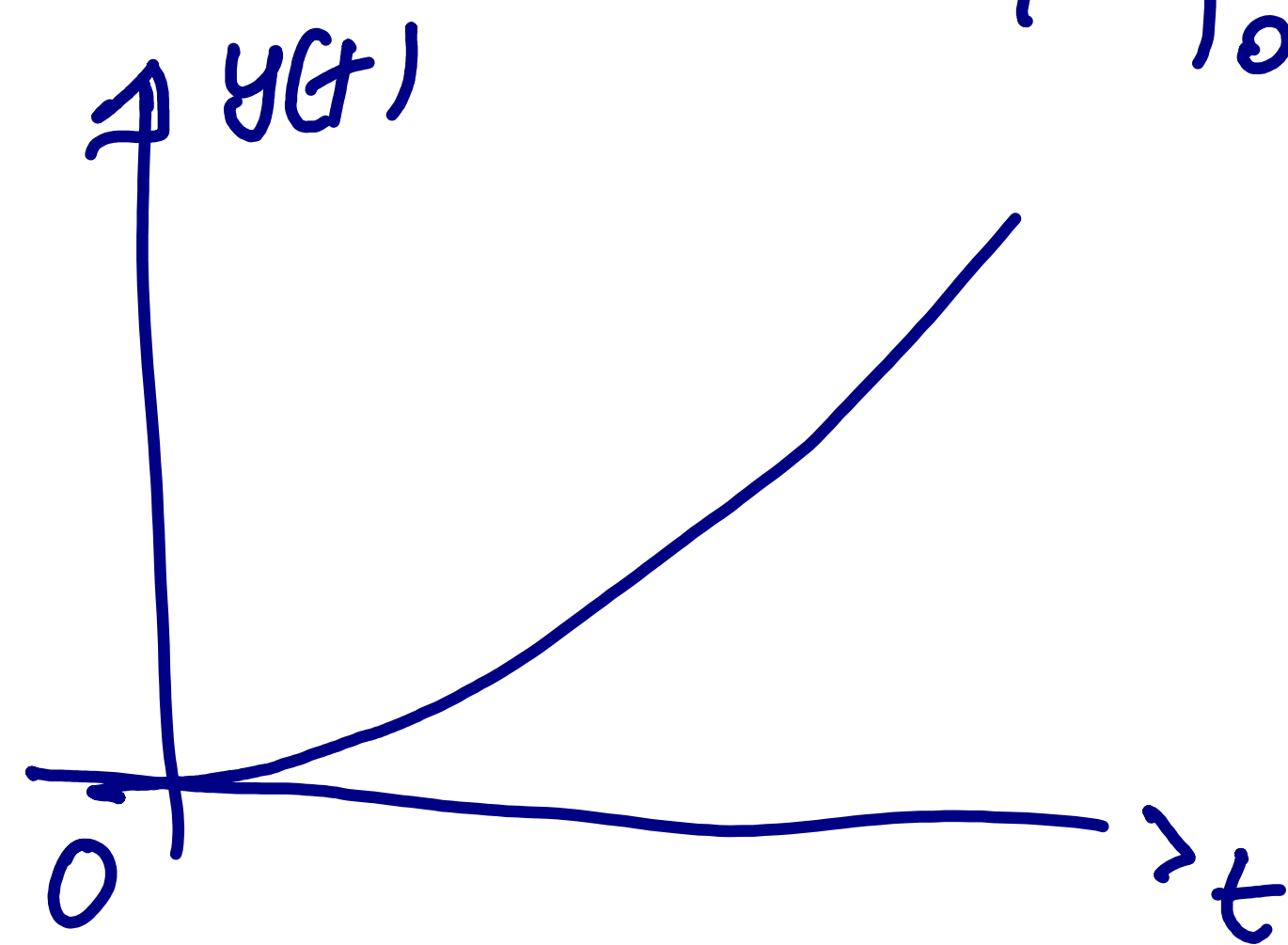
$$y(t) = \frac{1}{2} - \frac{1}{2}e^{-2t}; t \geq 0$$

$$y_d(t) = \frac{dy(t)}{dt} = \frac{K}{T}e^{-\frac{1}{T}t}; t \geq 0$$



$$y_t(t) = \int_0^t y_{1+}(\tau) d\tau = \int_0^t (k - k e^{-\frac{\tau}{T}}) d\tau = k\tau \Big|_0^t + k \frac{e^{-\frac{\tau}{T}}}{-\frac{1}{T}} \Big|_0^t =$$

$$= kt + kTe^{-\frac{t}{T}} - kT ; t \geq 0$$



Răspunsul sistemului de ord. II:

$$T_1 \frac{d^2 y(t)}{dt^2} + T_2 \frac{dy(t)}{dt} + y(t) = k u(t)$$

$$T^2 \frac{d^2 y(t)}{dt^2} + 2T \zeta \frac{dy(t)}{dt} + y(t) = k u(t)$$

$\nwarrow$ rap. de amortizare

$$T = \frac{1}{\omega_n} [\text{rad/s}]$$

$\nwarrow$ const. neamortizată

$$\frac{1}{\omega_n^2} \frac{d^2 y(t)}{dt^2} + 2 \frac{\zeta}{\omega_n} \frac{dy(t)}{dt} + y(t) = k u(t) \quad | \cdot \omega_n^2$$

$$\frac{d^2 y(t)}{dt^2} + 2\omega_n \zeta \frac{dy(t)}{dt} + \omega_n^2 y(t) = \omega_n^2 k u(t)$$

$$s^2 Y(s) + 2\zeta \omega_n s Y(s) + \omega_n^2 Y(s) = k \omega_n^2 U(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{k \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$y(t) = \mathcal{L}^{-1} \{ G(s) R(s) \}$$

Răsp. sistemului de ord. II depinde exclusiv de factorul de amortizare ξ .

Când $\xi = 0$, răsp. sistemului este oscilant amortizat. Când $\xi \in (0, 1)$, răsp. sistemului este subamortizat. Când $\xi = 1$, răsp. sistemului este critic amortizat, iar când $\xi > 1$, răsp. sistemului este supra-amortizat.

$$\xi \in (0, 1)$$

$$G_o(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\Delta = 4\xi^2\omega_n^2 - 4\omega_n^2 = -4\omega_n^2(1 - \xi^2)$$

$$p_{1,2} = \frac{-2\xi\omega_n \pm 2j\omega_n\sqrt{1-\xi^2}}{2} = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

$$G_o(s) = \frac{\omega_n^2}{(s + \xi\omega_n + j\omega_n\sqrt{1-\xi^2})(s + \xi\omega_n - j\omega_n\sqrt{1-\xi^2})} =$$

$$= \frac{\omega_n^2}{(s + \xi\omega_n)^2 + (\omega_n\sqrt{1-\xi^2})^2}$$

$$y(t) = \mathcal{L}^{-1}\{R(s)G_o(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s} \frac{\omega_n^2}{(s + \xi\omega_n)^2 + (\omega_n\sqrt{1-\xi^2})^2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{C_1}{s} + \frac{C_2 s + C_3}{(s + \xi\omega_n)^2 + (\omega_n\sqrt{1-\xi^2})^2}\right\}$$

$$C_1 = 1 \quad C_2 = -1 \quad C_3 = -2\xi\omega_n$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1-\zeta^2})^2} \right\}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1-\zeta^2})^2} - \frac{\zeta\omega_n\sqrt{1-\zeta^2} \frac{1}{\sqrt{1-\zeta^2}}}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1-\zeta^2})^2} \right\}$$

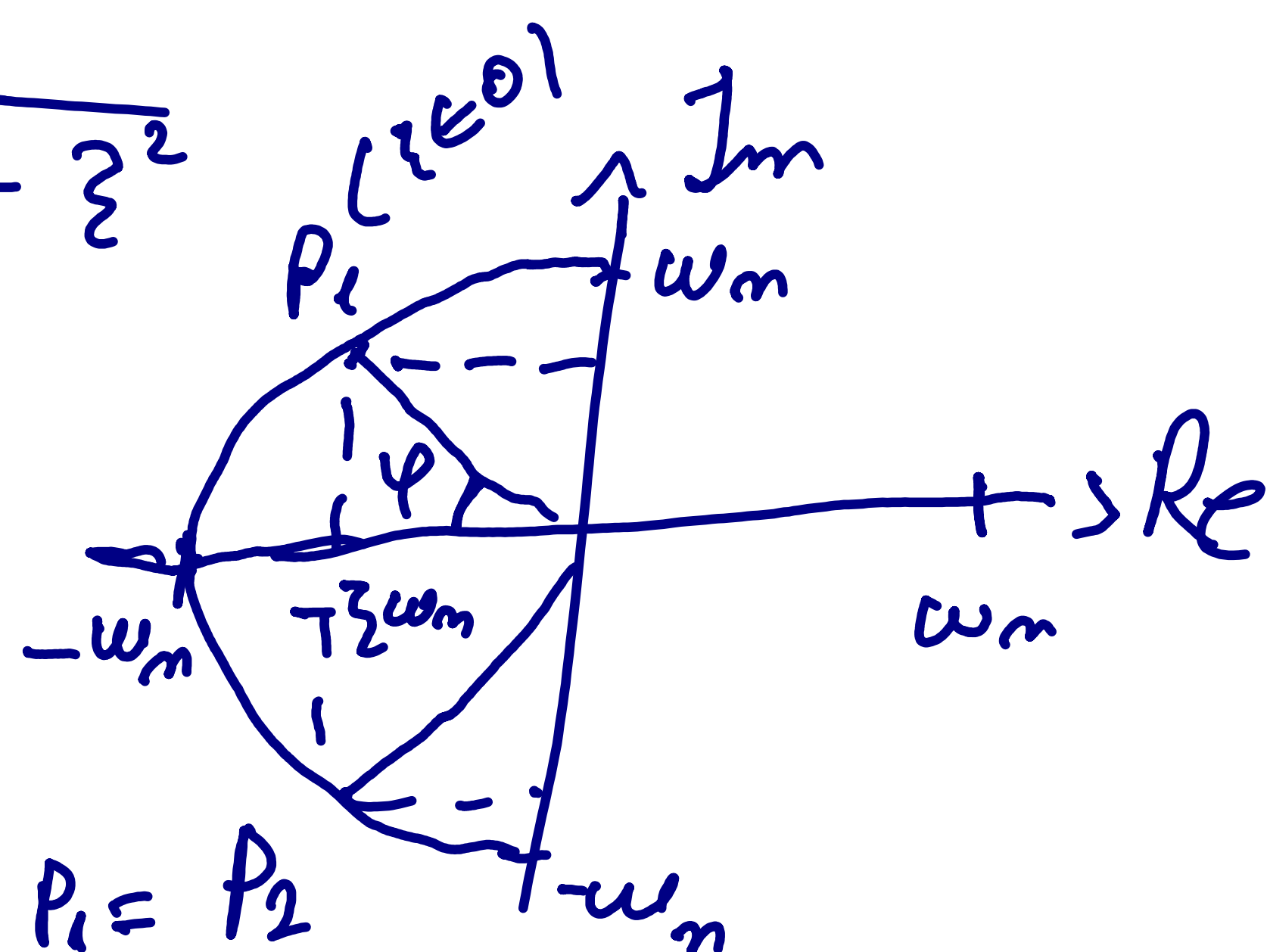
$$\Rightarrow y(t) = 1 - e^{-\zeta\omega_n t} \cos(\omega_n\sqrt{1-\zeta^2}t) - \frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n\sqrt{1-\zeta^2}t)$$

$$= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \left(\sqrt{1-\zeta^2} \cos(\omega_n\sqrt{1-\zeta^2}t) + \zeta \sin(\omega_n\sqrt{1-\zeta^2}t) \right)$$

$$\zeta \in (0, 1), p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$\zeta = 0, p_{1,2} = \pm j\omega_n$$

$$\zeta = 1, p_{1,2} = -\omega_n$$



$$\sin \varphi = \sqrt{1-\zeta^2}$$

$$\cos \varphi = \zeta$$

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n\sqrt{1-\zeta^2}t + \varphi)$$

